Chapter 5: Project Scope Management

Information Technology Project Management, Ninth Edition

Note: See the text itself for full citations

Twittie Senivongse (ed.)

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Learning Objectives (1 of 2)

- List key reasons why good project scope management is important
- Describe the process of planning scope management
- Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations
- Explain the scope definition process and describe the contents of a project scope statement
- Discuss the process for creating a work breakdown structure using the analogy, top-down, and bottom-up approaches
- Explain the importance of validating scope and how it relates to defining and controlling scope

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Learning Objectives (2 of 2)

- Given an information technology (IT) project situation, show how recommended approaches for controlling scope can improve the potential for project success
- Describe how software can assist in project scope management
- Discuss considerations for agile/adaptive environments

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

What is Project Scope Management?

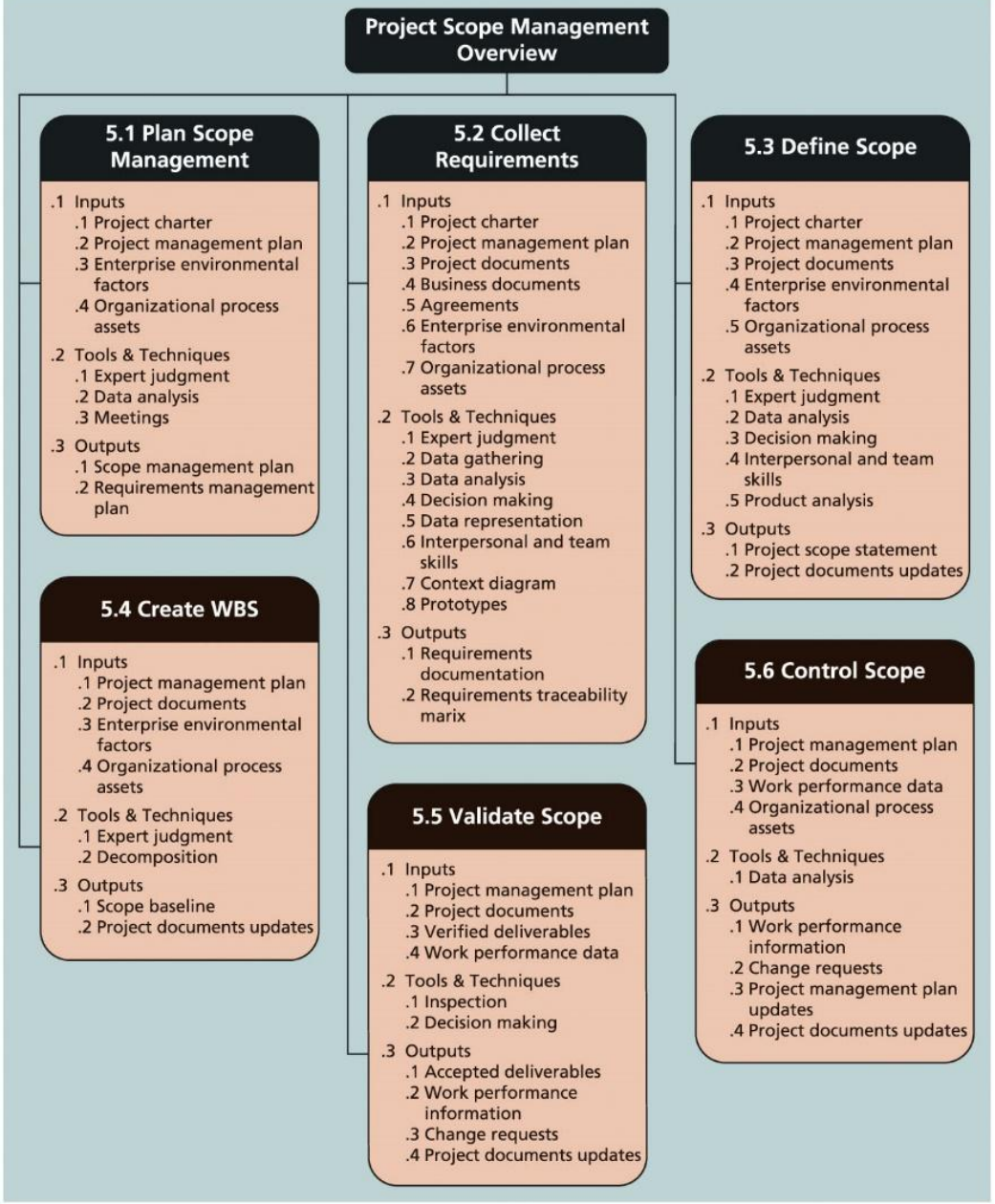
- Scope refers to all the work involved in creating the products of the project and the processes used to create them
  - A deliverable is a product produced as part of a project
    - Product related, e.g. hardware, software
    - Process related, e.g. planning documents, meeting minutes
- Project scope management includes the processes involved in defining and controlling what is or is not included in a project
  - Ensures that the project team and stakeholders have the same understanding of what products the project will produce and what processes the project team will use to produce them

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Project Scope Management Processes

- Main processes
  1. Planning scope management
     - Determining how the project's scope and requirements will be managed
  2. Collecting requirements
     - Defining and documenting the features and functions of the products produced during the project as well as the processes used for creating them
  3. Defining scope
     - Reviewing the project charter, requirements documents, and organizational process assets to create a scope statement
  4. Creating the WBS
     - Subdividing the major project deliverables into smaller, more manageable components, i.e. work breakdown structure
  5. Validating scope
     - Formalizing acceptance of the project deliverables
  6. Controlling scope
     - Controlling changes to project scope throughout the life of the project

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Source: *PMBOK® Guide – Sixth Edition*. Project Management Institute, Inc. (2017). Copyright and all rights reserved. Material from this publication has been reproduced with permission of PMI.

FIGURE 5-1 Project scope management overview

1. Planning Scope Management (1 of 2)

- The project team uses expert judgment, data analysis, and meetings to develop two important outputs
  - Scope management plan - subsidiary part of the project management plan (see chapter 4 project integration management)
  - Requirements management plan
- Small projects may not need written documents of such plans, but large projects may benefit from ones
- Scope management plan contents
  - How to prepare a detailed project scope statement – templates, guidelines
  - How to create a WBS – suggestions, samples
  - How to maintain and approve the WBS - guidelines
  - How to obtain formal acceptance of the completed project deliverables – process to obtain
  - How to control requests for changes to the project scope – guidelines (related to integrated change control in chapter 4 project integration management)

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

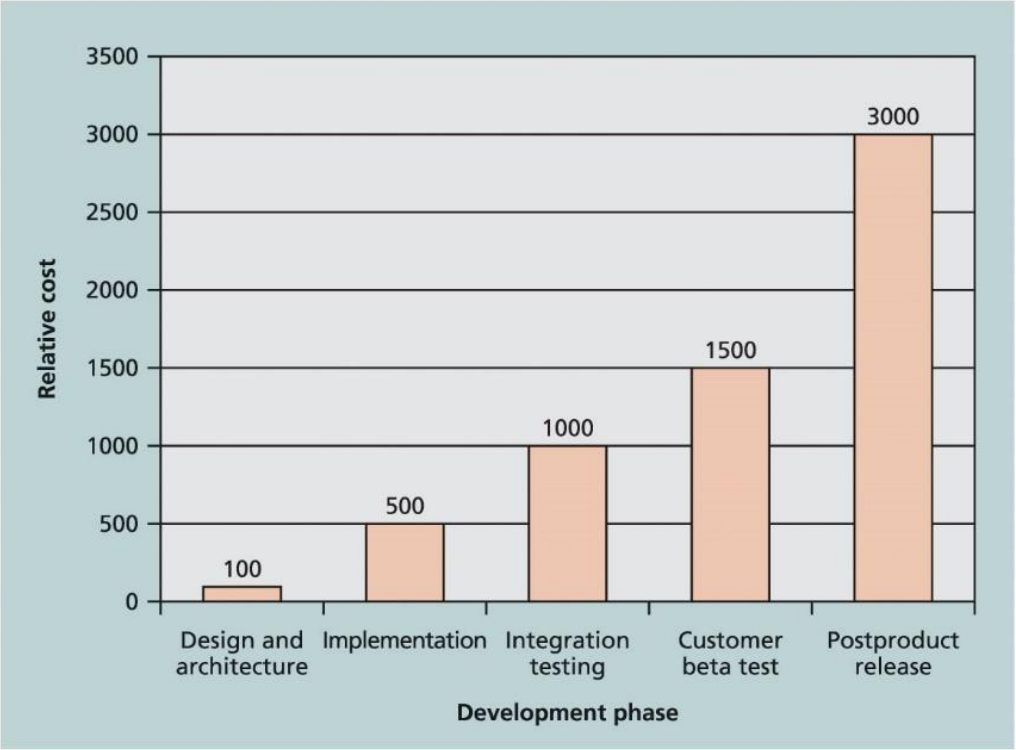
Planning Scope Management (2 of 2)

- A requirement is "a condition or capability that is necessary to be present in a product, service, or result to satisfy a business need" (PMBOK® Guide, Sixth Edition, 2017)
- The requirements management plan documents how project requirements will be analyzed, documented, and managed
  - How to plan, track, and report requirements activities
  - How to perform configuration management activities
  - How to prioritize requirements
  - How to use product metrics
  - How to trace and capture attributes of requirements

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

2. Collecting Requirements (1 of 3)

- Not defining requirements well causes rework which costs more when fixed in development phases later than requirement phase



Source: IBM Software Group, "Minimizing code defects to improve software quality and lower development costs," Rational Software (October 2008).

FIGURE 5-2 Relative cost to correct a software defect

Information Technol for use as permitted i part, except se.

Collecting Requirements (2 of 3)

- Several ways to collect requirements
  - Interviewing stakeholders
  - Holding focus groups and facilitated workshops
  - Using group creativity and decision-making techniques
  - Utilizing questionnaires and surveys
  - Conducting observation studies
  - Document analysis
  - Prototyping
  - Benchmarking (by comparing specific project practices or product characteristics to others inside or outside organization)

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Collecting Requirements (3 of 3)

- Requirements documentation
  - Text, images, diagrams, video, other media
  - Often broken down into categories, e.g. functional, nonfunctional, service, quality, training requirements
- Requirements traceability matrix (RTM)
  - Table that lists requirements, each with various attributes and status, to ensure that all requirements are addressed

Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet memory requirement.

Table 5-1 Sample entry in a requirements traceability matrix of an IT upgrade project

  - In software development, RTM maintains linkage from source of each requirement to implementation and testing

https://www.opencodez.com/software-testing/create-requirement-traceability-matrix-rtm-free-sample-download.htm

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

3. Defining Scope (1 of 2)

- Important elements of a project scope statement
  - Product scope description
  - Product user acceptance criteria
  - Detailed information on all project deliverables
  - Other scope-related information
    - Project boundaries, constraints, and assumptions
    - Supporting document references (e.g. product specifications)
  - See example in chapter 3
- As time progresses, the scope of a project should become more clear and specific

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Defining Scope (2 of 2)

Project Charter:
Upgrades may affect servers . . . (listed under Project Objectives)

Project Scope Statement, Version 1:
Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.

Project Scope Statement, Version 2:
Servers: This project will require purchasing 10 new servers to support Web, network, database, application, and printing functions. Virtualization will be used to maximize efficiency. Detailed descriptions of the servers are provided in a product brochure in Attachment 8, along with a plan describing where they will be located.

Table 5-3 Further defining project scope

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

4. Creating the Work Breakdown Structure (1 of 10)

- Work Breakdown Structure (WBS) is a deliverable-oriented grouping of the work involved in a project that defines the total scope of the project
  - Foundation document that provides the basis for planning and managing project schedules, costs, resources, and changes
- Decomposition is the main tool or technique for creating a WBS
  - Subdividing project deliverables into smaller pieces
  - A work package is a task at the lowest level of the WBS
- PMI recommends using nouns to name WBS items
  - Activities below use verbs (see chapter 6 project schedule management)
- Outputs of creating the WBS are the scope baseline and project documents updates
  - Scope baseline includes the approved project scope statement and its associated WBS and WBS dictionary

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Creating the Work Breakdown Structure (2 of 10)

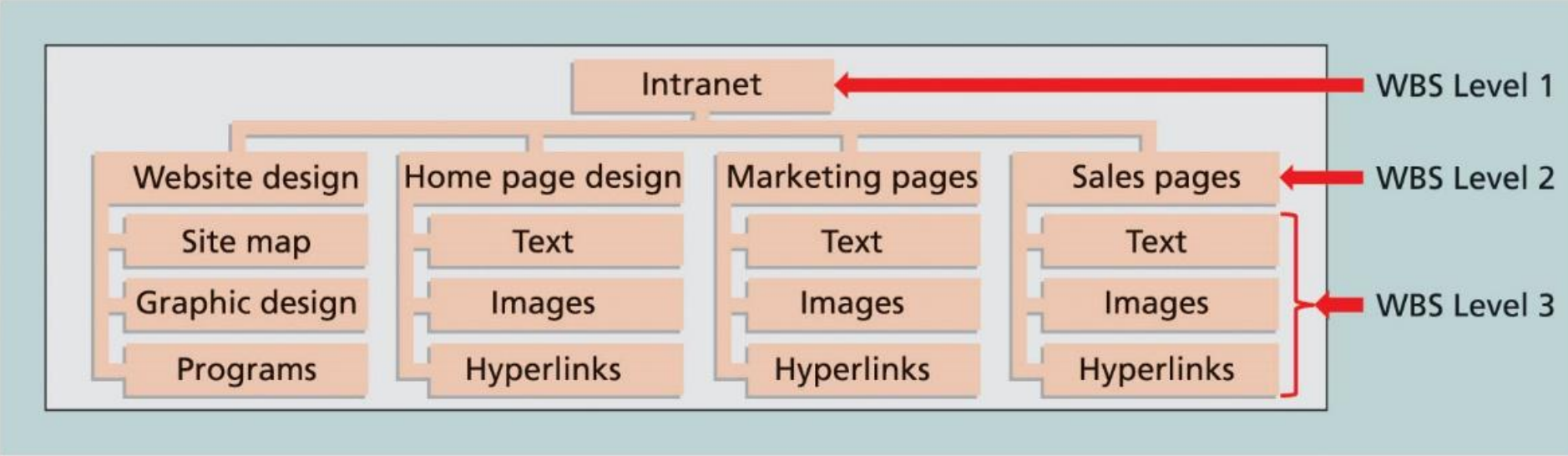
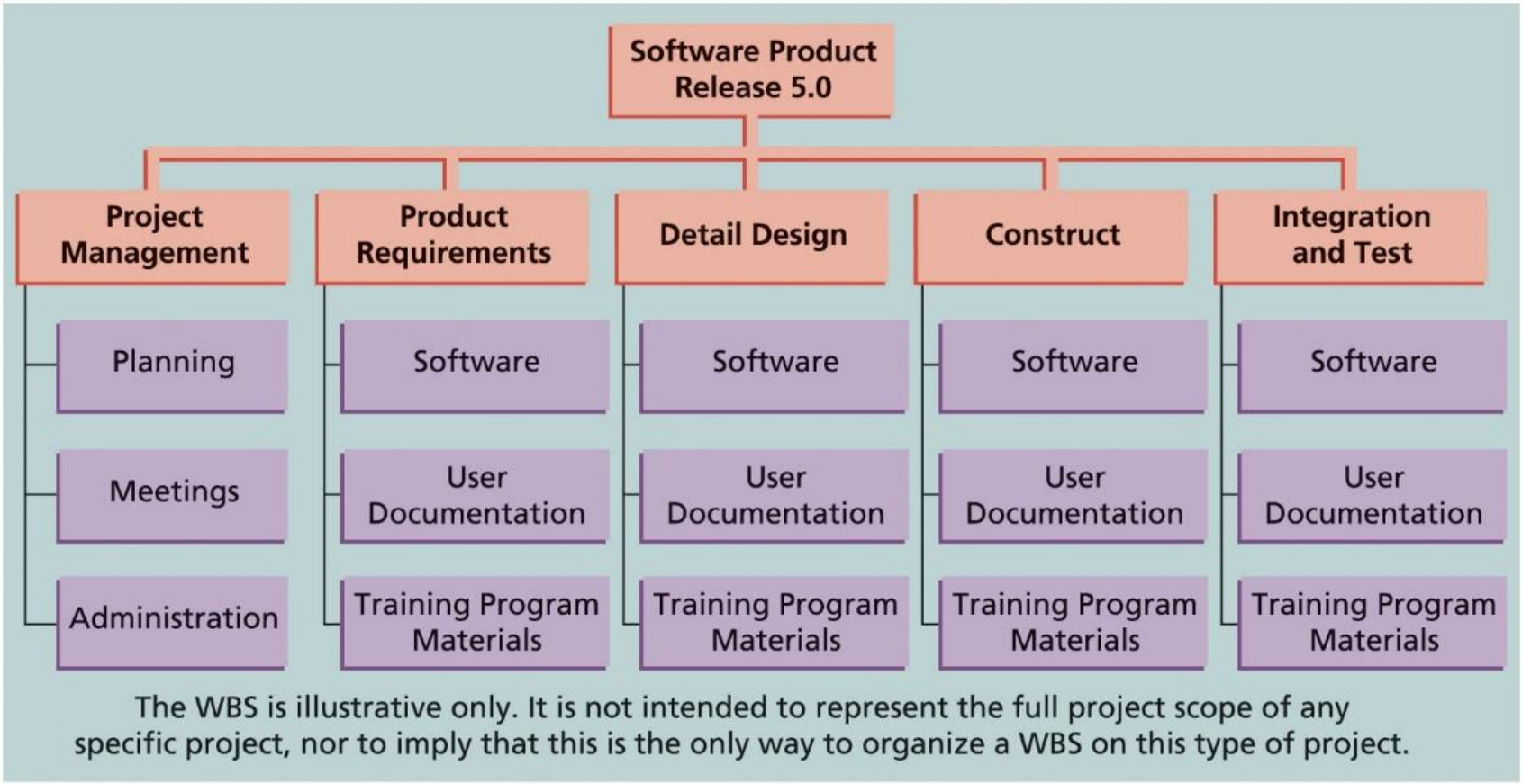


FIGURE 5-3 Sample intranet WBS organized by product

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Creating the Work Breakdown Structure (3 of 10)



Source: *PMBOK® Guide – Sixth Edition*. Project Management Institute, Inc. (2017). Copyright and all rights reserved. Material from this publication has been reproduced with permission of PMI.

FIGURE 5-4 Sample intranet WBS organized by phase in chart and tabular form

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Creating the Work Breakdown Structure (4 of 10)

1.0 Software Product Release 5.0		
	1.1 Project Management	
		1.1.1 Planning
		1.1.2 Meetings
		1.1.3 Administration
	1.2 Product Requirements	
		1.2.1 Software
		1.2.2 User Documentation
		1.2.3 Training Program Materials
	1.3 Detail Design	
		1.3.1 Software
		1.3.2 User Documentation
		1.3.2 User Documentation
	1.4 Construct	
		1.4.1 Software
		1.4.2 User Documentation
		1.4.3 Training Program Materials
	1.5 Integration and Test	
		1.5.1 Software
		1.5.2 User Documentation
		1.5.3 Training Program Materials

Table 5-4 Tabular form of WBS

Creating the Work Breakdown Structure (5 of 10)

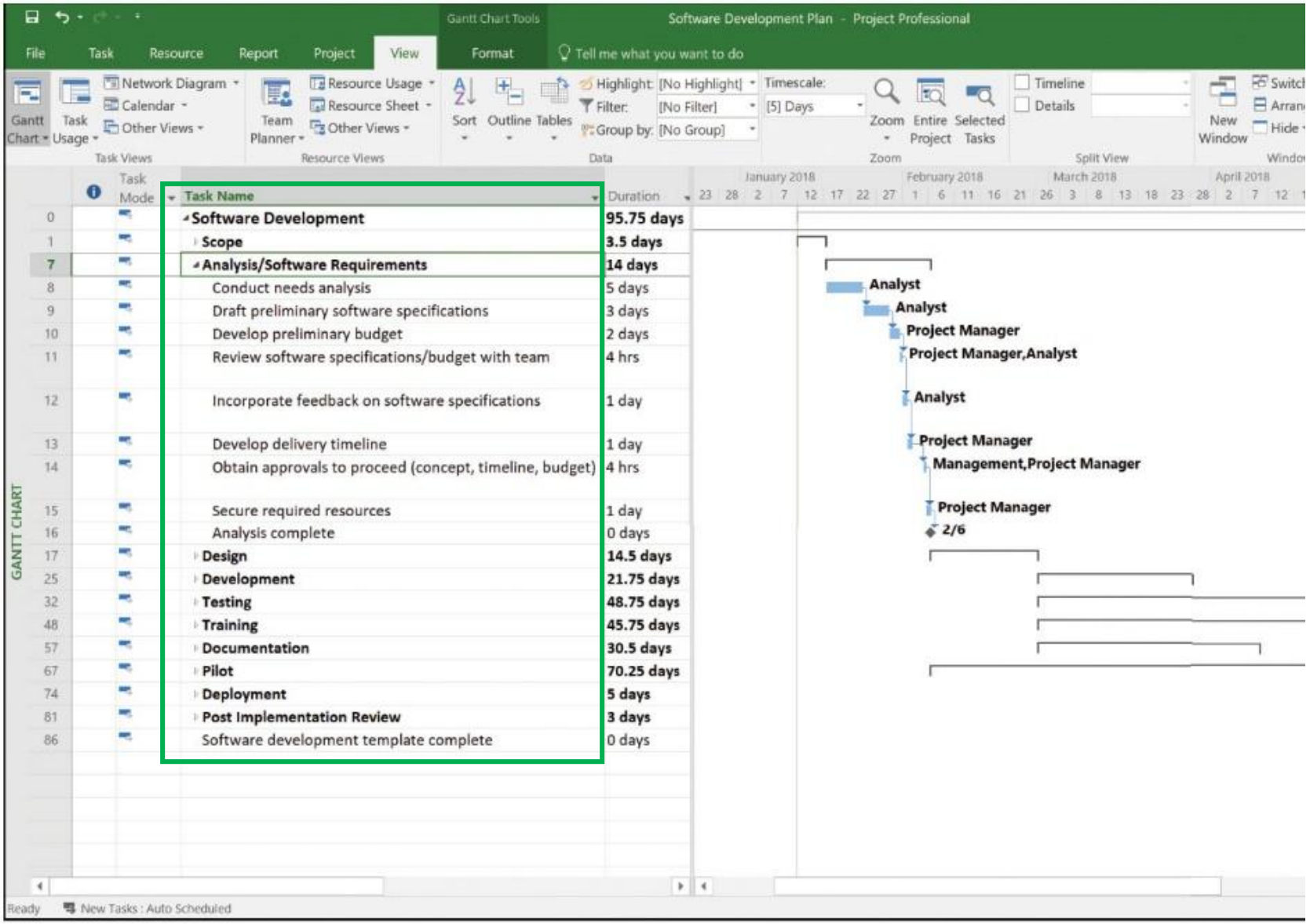
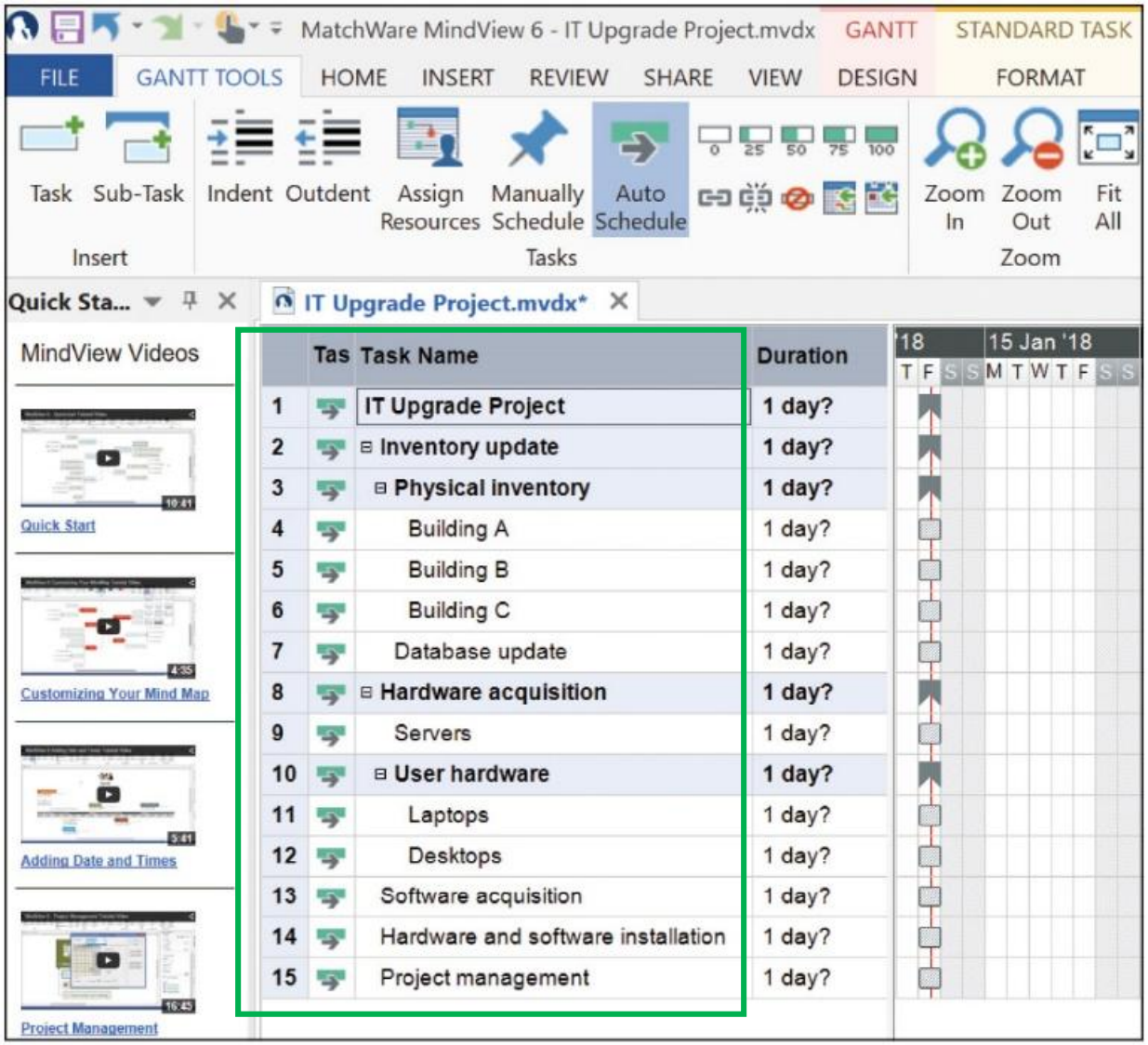


FIGURE 5-5 Software development project template from Microsoft Project 2016

Creating the Work Breakdown Structure (6 of 10)



Source: MatchWare MindView 6.0

FIGURE 5-8 Gantt chart with WBS generated from a mind map

Creating the Work Breakdown Structure (7 of 10)

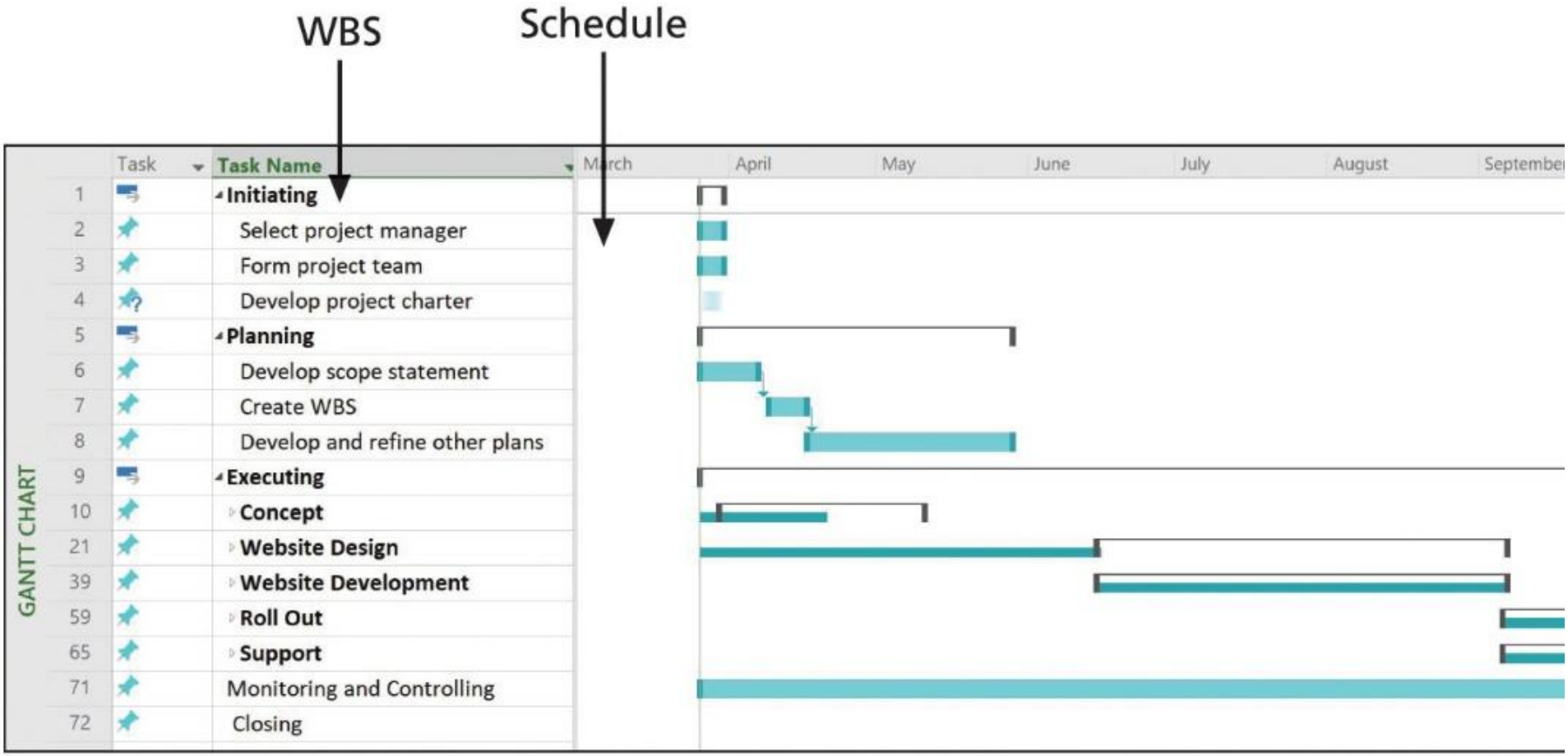


FIGURE 5-6 Website project Gantt chart organized by project management process groups

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Creating the Work Breakdown Structure (8 of 10)

WBS by phase (i.e. process decomposition) in tabular form of a Scrum project (Ride Sharing)

Process decomposition

Task Number	Task Name	Duration (days)	Dependency	Status
1	Initiating			
1.1	Develop a Business Case	5		Complete
1.2	Undertake a Feasibility Study	2		Complete
1.3	Establish the Project Charter	1	1.2	Complete
1.4	Create the Project Team and Determine Roles	1	1.3	Complete
1.5	Determine Total Number of Sprints and Releases	3	1.4	Complete
1.6	Kick-Off Meeting	1	1.5	Complete
2	Planning			
2.1	Create Product Backlog and Sprint Backlog	7	1.6	Open
2.2	Create Document for Sprint Planning	5	2.1	Open
2.3	Create a Resource Plan	3	2.2	Open
2.4	Create a Financial Plan	5	2.3	Open
2.5	Create a Budget Plan	3	2.4	Open
3	Development			
3.1	Requirement Gathering			
3.1.1	Identify User's need and Survey	5	2.5	Open
3.2	Training			
3.2.1	Knowledge Sharing Session	2	3.1.1	Open
3.2.2	Technology and Tools Session	2	3.2.1	Open
3.2.3	Setup Session	4	3.2.2	Open
3.2.4	Existing System Session	2	3.2.3	Open
3.3	Executing (System & Testing)			
3.3.1	Sprint 1			
3.3.1.1	Sprint Planning Route System, Meter Calculator System	1	3.2.4	Open
3.3.1.2	Daily Scrums	14	3.2.4	Open
3.3.1.3	Implement	11	3.3.1.1	Open
3.3.1.4	Testing	8	3.3.1.3	Open
3.3.1.5	Sprint Review 1	1	3.3.1.4	Open
3.3.1.6	Sprint Retrospective 1	1	3.3.1.5	Open
3.3.2	Sprint 2			
3.3.2.1	Sprint Planning Accounting Management System	1	3.3.1.6	Open
3.3.2.2	Daily Scrums	14	3.3.1.6	Open
3.3.2.3	Implement	11	3.3.2.1	Open
3.3.2.4	Testing	8	3.3.2.3	Open
3.3.2.5	Sprint Review 2	1	3.3.2.4	Open
3.3.2.6	Sprint Retrospective 2	1	3.3.2.5	Open
3.3.3	Sprint 3			
3.3.3.1	Sprint Planning Messaging System	1	3.3.2.6	Open
3.3.3.2	Daily Scrums	14	3.3.2.6	Open
3.3.3.3	Implement	11	3.3.3.1	Open
3.3.3.4	Testing	8	3.3.3.3	Open
3.3.3.5	Sprint Review 3	1	3.3.3.4	Open
3.3.3.6	Sprint Retrospective 3	1	3.3.3.5	Open
3.3.4	Sprint 4			
3.3.4.1	Sprint Planning Route Notification, User Notification	1	3.3.3.6	Open
3.3.4.2	Daily Scrums	14	3.3.3.6	Open
3.3.4.3	Implement	11	3.3.4.1	Open
3.3.4.4	Testing	8	3.3.4.3	Open
3.3.4.5	Sprint Review 4	1	3.3.4.4	Open
3.3.4.6	Sprint Retrospective 4	1	3.3.4.5	Open
3.3.5	Software Release	2	3.3.4.6	Open
3.4	Documentation			
3.4.1	User's Guide	4	3.3.5	Open
3.4.2	Installation Guide	1	3.4.1	Open
4	Monitoring and Controlling			
4.1	Perform Burndown Chart	71		Open
4.2	Perform Cost Management	71		Open
4.3	Perform Risk Management	71		Open
4.4	Perform Change Management	71		Open
5	Closing			
5.1	Review Project Completion	3	3.4.2	Open

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Creating the Work Breakdown Structure (9 of 10)

Zoom in on planning product backlog

WBS by product decomposition

Product Backlog

EPIC ID	EPIC Name	User Story ID	User Story	Estimate (User Story Points)
EPIC 1	Matching system	US1-1	As a patient I want to select a physician so that I can consult a physician	13
		US1-2	As a physician I want to accept patients' request to consult with me so that I can give consultations to patients	13
		US1-3	As a physician I want to decline patients' requests so that the patients can send requests to another physician	13
		US1-4	As a patient I want to be able to sort the physician's list by the relevant factors so that I can select the best physicians for me	3
		US1-5	As a patient I want to see the physician's working hours so that I can select the best physicians that are available	2
		US1-6	As a patient I want to search physicians so that I can select appropriate physicians	5
		US1-7	As a patient I want to see the matching status so that I can know where my request state is in	3
		US1-8	As a patient I want to cancel the request so that I am free to change my plan	2
EPIC 2	Registration and Login system	US2-1	As a patient I want to register to the system so that I can consult a physician	8
		US2-2	As a physician I want to register to the system so that I can give consultations to patients	8
		US2-3	As a patient I want to provide a credit/debit card information so that I can pay service fees	3
		US2-4	As a physician I want to provide my Medical License, specialties and working hour so that I will be reliable and credible	3

Informat
use as p
ot for

Creating the Work Breakdown Structure (10 of 10)

- Approaches to developing work breakdown structures
  - Using guidelines
    - Some organizations, like the U.S. Department of Defense (DOD), provide guidelines for preparing WBSs
  - Analogy approach
    - Review WBSs of similar projects and tailor to your project
  - Top-down approach
    - Start with the largest items of the project and break them down
  - Bottom-up approach
    - Start with the specific tasks and aggregate them into higher levels

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

WBS Dictionary (1 of 2)

- Many WBS tasks are vague
- WBS dictionary is a document that describes detailed information about each WBS item
  - Format of the WBS dictionary can vary based on project needs

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

WBS Dictionary (2 of 2)

WBS Dictionary Entry March 20
Project Title: Information Technology (IT) Upgrade Project
WBS Item Number: 2.2
WBS Item Name: Database Update
Description: The IT department maintains an online database of hardware and software on the corporate intranet. We need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Physical Inventory, and must precede WBS Item Number 3.0, Hardware and Software Acquisition.

Table 5-5 Sample WBS dictionary entry

Advice for creating a WBS and WBS dictionary

- Unit of work should appear at only one place in the WBS
- Work content of a WBS item is the sum of the WBS items below it
- WBS item is the responsibility of only one individual, even though many people may be working on it
- WBS must be consistent with the way in which work is actually going to be performed
- Project team members should be involved in developing the WBS to ensure consistency and buy-in
- Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

5. Validating Scope

- Scope validation involves formal acceptance of the completed project deliverables
  - Acceptance is often achieved by a customer inspection and then sign-off on key deliverables
  - Sponsor or user inspects work after it is delivered and decides if it meets requirements.
    - Outputs can be accepted deliverables or change requests

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

6. Controlling Scope (1 of 3)

- Scope control involves controlling changes to the project scope
  - Keeping project goals and business strategy in mind
- Even when the project scope is fairly well defined, many IT projects suffer from scope creep
  - Tendency for project scope to keep getting bigger and bigger
- Goals of scope control
  - Influence the factors that cause scope changes
  - Ensure changes are processed according to procedures developed as part of integrated change control
  - Manage changes when they occur
- Variance analysis is a tool for scope control
  - Variance is the difference between planned and actual performance
  - E.g. If delivering four functions when it should be five, variance is one function



Project Scope Creep https://www.youtube.com/watch?v=1e1tkmg-bXQ&t=42s

Dealing With Scope Creep https://www.youtube.com/watch?v=PAdXJLtEVPE

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Controlling Scope (2 of 3)

- Suggestions for improving user input
  - Develop a good project selection process and insist that sponsors are from the user organization
  - Place users on the project team
  - Conduct regular meetings with defined agendas
  - Deliver something to users and sponsors on a regular basis
  - Do not promise to deliver what the team cannot deliver in a particular time frame
  - Locate users with the developers

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Controlling Scope (3 of 3)

- Suggestions for reducing incomplete and changing requirements
  - Develop and follow a requirements management process
  - Employ techniques such as prototyping, use case modeling, and Joint Application Development (JAD) to get more user involvement
  - Put requirements in writing and keep them current
  - Create a requirements management database for documenting and controlling requirements
  - Provide adequate testing and conduct it throughout the project life cycle
  - Review changes from a systems perspective
  - Emphasize completion dates to help focus on what's most important
  - Allocate resources specifically for handling change requests

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Using Software to Assist in Project Scope Management

- Word-processing software helps create several scope-related documents
- Spreadsheets or presentation software to develop various charts, graphs, and matrixes related to scope management
- Communication software like e-mail and web-based applications can transmit project scope management information
- Project management software helps in creating a WBS; basis for creating a Gantt chart
- Specialized software is available to assist in project scope management, e.g. requirements management software









Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Considerations for Agile/Adaptive Environments

- Stakeholders define and approve the detailed scope before the start of an iteration with an adaptive or agile product life cycle, producing a usable product at the end of each iteration
  - Detailed scope develops over time
- Agile approach provides several usable products during the project

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Chapter Summary

- Project scope management includes the processes required to ensure that the project addresses all the work required, and only the work required, to complete the project successfully
  - Main processes
    - Define scope management
    - Collect requirements
    - Define scope
    - Create WBS
    - Validate scope
    - Control scope

Information Technology Project Management, Ninth Edition. © 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.